

VISHAL BHARTI

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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate and aspiring Software Development Engineer with hands-on experience in backend engineering, full-stack development, and AI-driven applications. Proficient in C++, Python, Data Structures & Algorithms, Object-Oriented Programming, SQL, and REST APIs. Built and shipped a Retrieval-Augmented Generation (RAG) system, full-stack web applications, and an RFID-based IoT system. AWS Certified Solutions Architect – Associate.

SKILLS

Programming Languages: C++, Python, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems

Web & Backend: HTML, CSS, JavaScript, Flask, REST APIs

Databases: MySQL

Cloud and Developer Tools: AWS, Git, GitHub, VS Code

IoT & Platforms: Arduino IDE, ThingSpeak

PROJECTS

Scalable Document Question Answering System (RAG) | Backend & AI Developer Apr 2026 - Present

- Engineered a Retrieval-Augmented Generation (RAG) pipeline in Python using LangChain and FAISS, enabling context-aware Q&A across multi-document corpora of 100+ pages.
- Optimized semantic search via vector embeddings and chunking strategy, improving retrieval relevance and reducing irrelevant context passed to the LLM.
- Architected a modular, scalable pipeline separating ingestion, retrieval, and generation stages, cutting the effort to add new document types to under an hour.

Root Finder Web Application | Full-Stack Developer Jan 2026 - Mar 2026

- Built and deployed a full-stack numerical computing app supporting 3+ root-finding algorithms (Bisection, Newton-Raphson, Secant) with real-time graph rendering via Plotly.
- Designed a Flask REST backend to handle computation requests and return real-time results, supporting concurrent root-finding queries with sub-second response times.
- Delivered a responsive HTML/CSS/JS frontend, tested across desktop and mobile viewports for consistent UX.

Smart Robotic Waiter System (IoT + RFID) | IoT Developer Sep 2025 - Dec 2025

- Built an autonomous robotic delivery system on ESP32 combining RFID table identification with obstacle-avoidance navigation, eliminating manual order-tracking steps entirely.
- Integrated real-time sensor data (RFID + proximity) with embedded control logic to automate table-to-table delivery, reducing average delivery time versus manual service.

EDUCATION

Chandigarh University — B.E. in Computer Science Engineering | Aug 2023 - Present

CGPA: 7.92

Sunbeam School — Intermediate (CBSE) | 2022 - 2023

Percentage: 68.40%

St. Francis School — Matriculation (CBSE) | 2020 - 2021

Percentage: 93.40%

ACHIEVEMENTS

- Advanced to Phase 3 of the University Project Expo with a technical project.
- Solved 200+ Problems on LeetCode

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (2025)
- Data Structures & Algorithms — Infosys Springboard (2025)
- Database & SQL — Infosys Springboard (2025)
- Master Python & Generative AI for Advanced Analytics — Udemy (2025)